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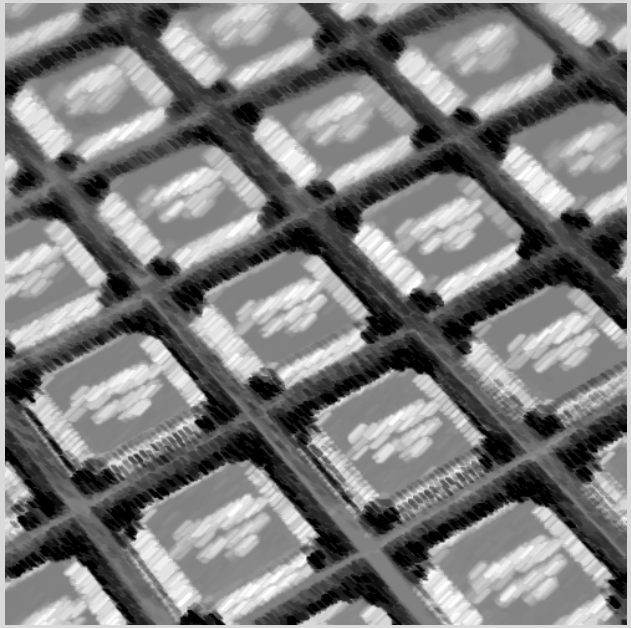
Embedded Systems

Architectures

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Università di Genova

ASICs and ASSPs



What are ASICs and ASSPs

ASIC (Application-Specific Integrated Circuit): an integrated circuit specifically designed for a special single purpose or application.

Main Characteristics:

- It is not a general-purpose device (application specific)
- ASIC is built for one and only one customer.
- The intellectual property, design database, and deployment of an ASIC are usually controlled by just a single entity (i.e. it is not available to the public.)

Examples:

- an IC that encodes and decodes digital data using a proprietary encoding/decoding algorithm;
- an amplifier IC designed to meet certain specifications not available in standard amplifier products;
- a medical IC designed for a machine monitoring a specific human biometric parameter;

What are ASICs and ASSPs

ASSP (Application-Specific Standard Product): an integrated circuit specifically designed for a special application market.

Main Characteristics:

- It is not a general-purpose device (application specific).
- ASSP is built for more than one customer (standard product)
- The intellectual property, design database, and deployment of an ASIC are available to a certain level

Examples:

- an image processing IC performing an operation like reducing image blurring.
- an IC for motor control
- an IC for wireless communication

What are ASICs and ASSPs

Characteristics of ASICs and ASSPs

- Fixed Hardware: Internal logic is fixed by manufacturers.
- No Programmable Software: The IC may include sequential elements (like microcontrollers) but their instructions are fixed (firmware) and not modifiable by user (apart from updates, eventually available from manufacturers)

Pros and Cons of ASICs and ASSPs

Main pros:

- Design optimized for the application:
  - number of gates tailored to the application needs
  - reduced power consumption
  - reduced IC size
  - improved performances
- Low costs for replica production (they just “contain” what needed)

Main cons:

- High design costs (hundreds of k€) for producing IC masks (an IC mask is a glass plate with a pattern of transparent and opaque areas used to photolithographically create patterns on wafers)
- Low flexibility
  - no option for circuit modification during fabrication life-cycle of an ASIC/ASSP linked with life-cycle of target products

Conclusion on ASICs and ASSPs

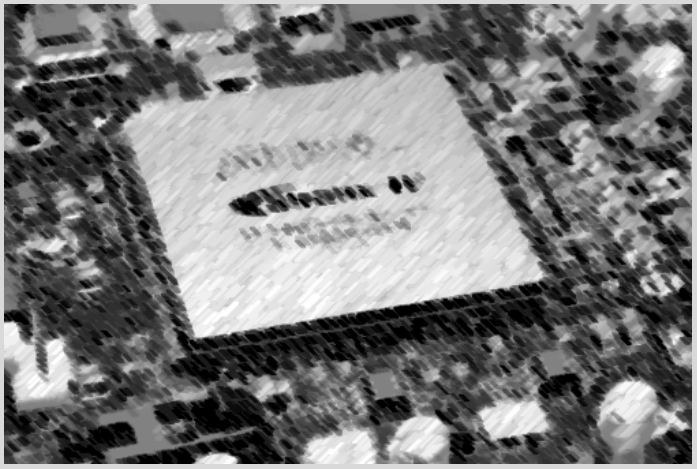
Fixed logic nonprogrammable devices (ASICs and ASSPs) are optimal solutions for high-volume productions. They can be found inside several objects of common use (consumer electronics, house appliances, ...)

For medium- or small-volume productions NRE (Non-Recurring Engineering) costs of fixed logic device are generally not sustainable.

For medium- or small-volume productions a better (i.e. more cost-effective) solution is using standard components, such as small sequential devices (like DSPs or microcontrollers) or hardware-reconfigurable devices (like PLDs or FPGAs).

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PLDs, CPLDs and FPGAs



What are PLDs

PLD (Programmable Logic Device): an integrated circuit whose internal logic is not rigid but can be configured to some extent according to the user needs.

Main Features:

- PLDs are standard products sold as COTS (commercial-off-the-shelf).
- A PLD has an undefined function at the time of manufacture and before it can be used in a circuit it must be programmed (i.e. internally configured).
- PLDs have been developed since 70s. Originally, they were used for implementing glue logics (i.e. custom electronic circuitries interfacing two or more different ICs).

What are PLDs

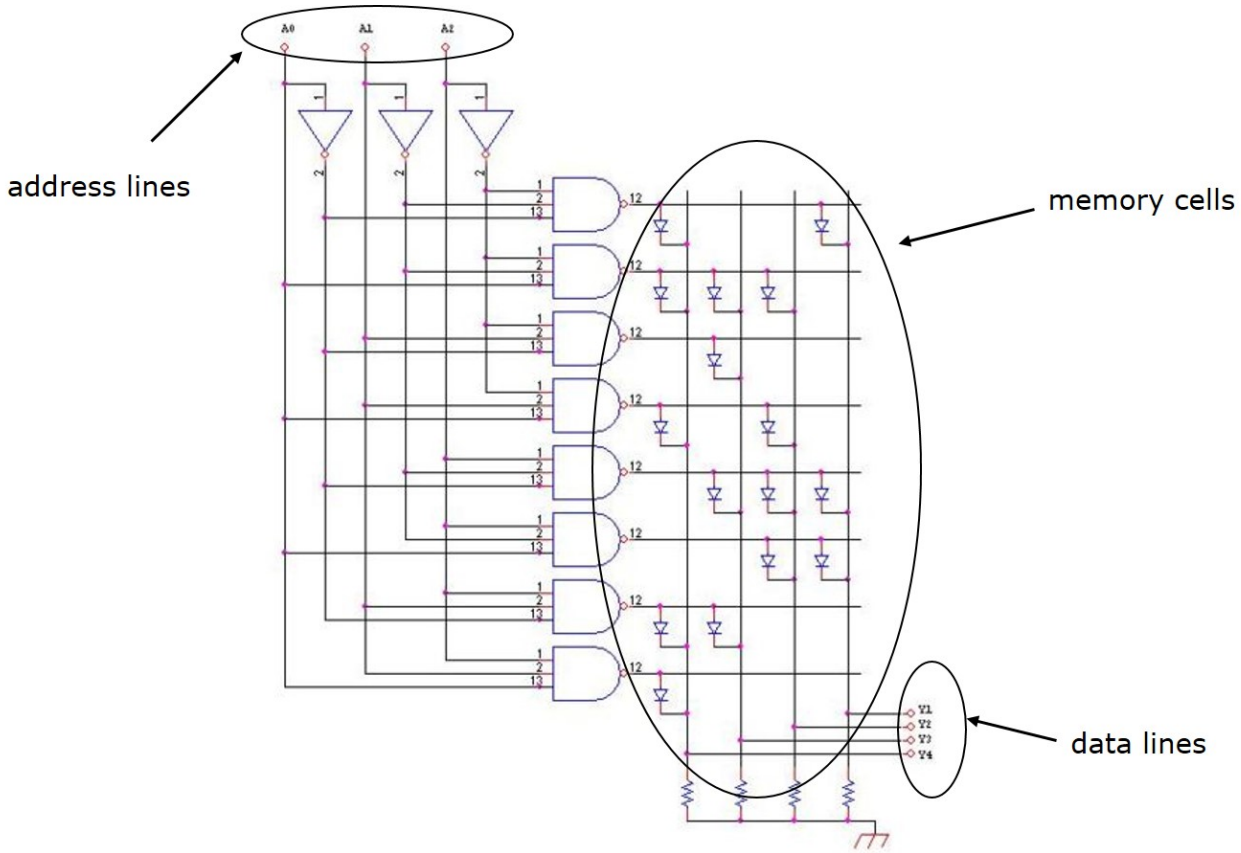
They are often preferred to sets of gates, since they allow to:

- reduce the number of employed components
- save PCB space
- save power consumption.

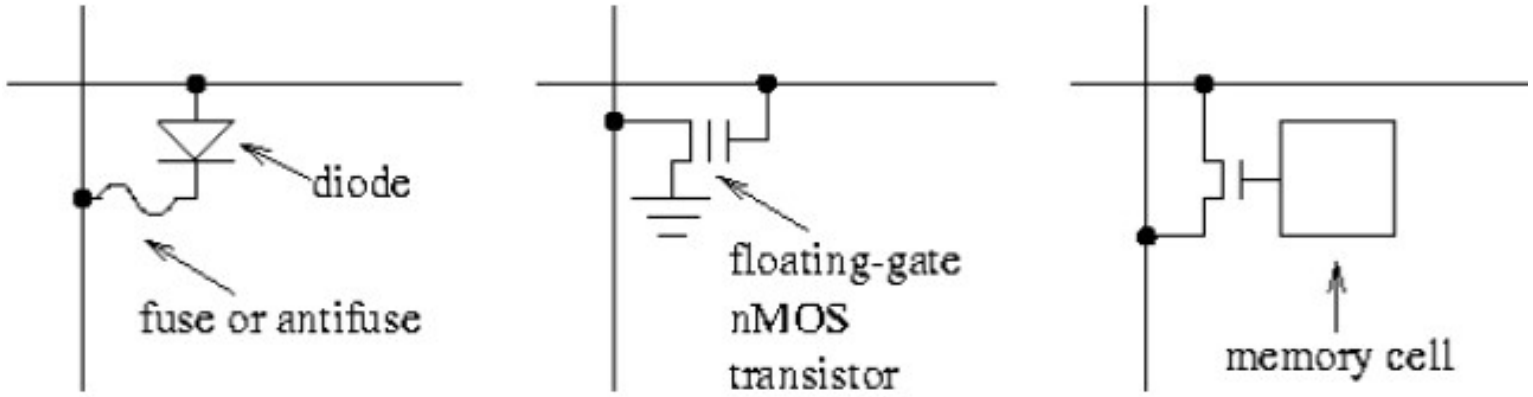
As feature sizes have shrunk and design tools improved over the years, functionalities of PLDs have sensibly increased.

A wide range of powerful reconfigurable devices is today employed in several medium- or low-volume productions.

PLD Example: a Simple PROM



PROM Technologies



OTP: One Time Programmable (fuse or antifuse)

MTP: Multiple Times Programmables (floating-gate transistor, SRAM, Flash)

(in-circuit programmable)

PROM Example

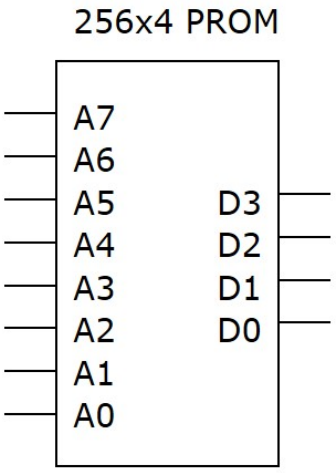
Example of application

Implement the following logic function

$f1 = \overline{w}x\overline{y}+\overline{x}y\overline{z}+xz$

$f2 = \overline{ab}y+ab\overline{xy}+\overline{a}b\overline{xy}$

by using a 256x4 PROM



Solution

First write down the truth table

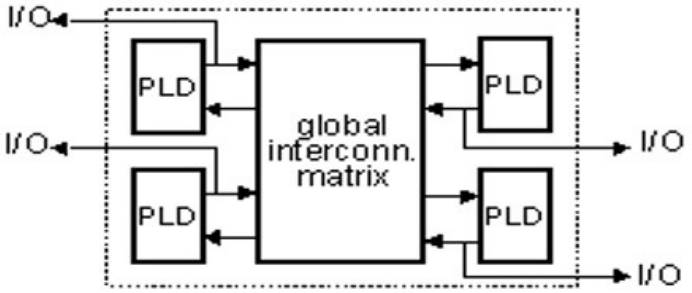
a	b	w	x	y	z	f1	f2
0	0	x	x	1	x	0	1
0	1	x	0	0	x	0	1
1	1	x	0	0	x	0	1
x	x	x	0	1	0	1	0
x	x	0	1	0	x	1	0
x	x	x	1	x	1	1	0
all others						0	0

Then map it to the chip I/O lines

A7	A6	A5	A4	A3	A2	A1	A0	D3	D2	D1	D0
a	b	w	x	y	z			f1	f2		
0	0	0	0	x	x	1	x	0	1	-	-
0	0	0	1	x	0	0	x	0	1	-	-
0	0	1	1	x	0	0	x	0	1	-	-
0	0	x	x	x	0	1	0	1	0	-	-
0	0	x	x	0	1	0	x	1	0	-	-
0	0	x	x	x	1	x	1	1	0	-	-
all others								0	0	-	-

What are CPLDs

CPLD (Complex PLD): a PLD made up of contains a bunch of PLD blocks whose inputs and outputs are connected by a global programmable interconnection matrix



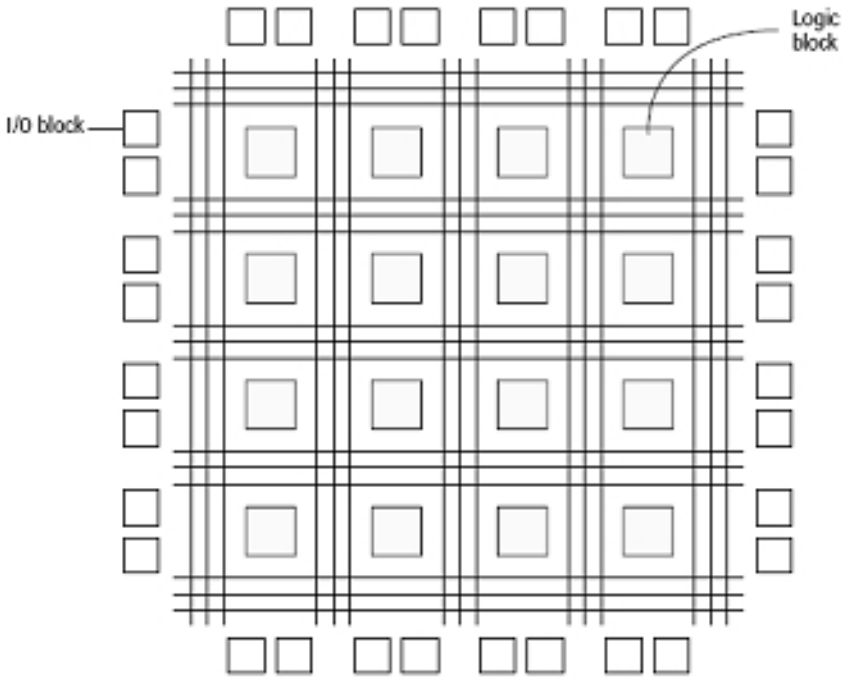
What are CPLDs

Main Features: Two levels of programmability

- each PLD block (logic block)
- the interconnections between the PLDs (Programmable Interconnection Matrix, PIM)

What are FGPAs

FPGA (Field Programmable Gate Array): a PLD made up of a bunch of simple, configurable logic and I/O blocks arranged in an array with interspersed switches that can rearrange the interconnections between the logic and I/O blocks.



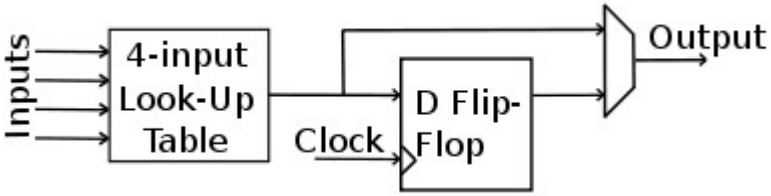
Main Features:

- FPGA has is a “fine-grain” device (each logic block can do little by itself)
- FPGA has two levels of programmability:
- each logic and I/O block
- the interconnections between blocks

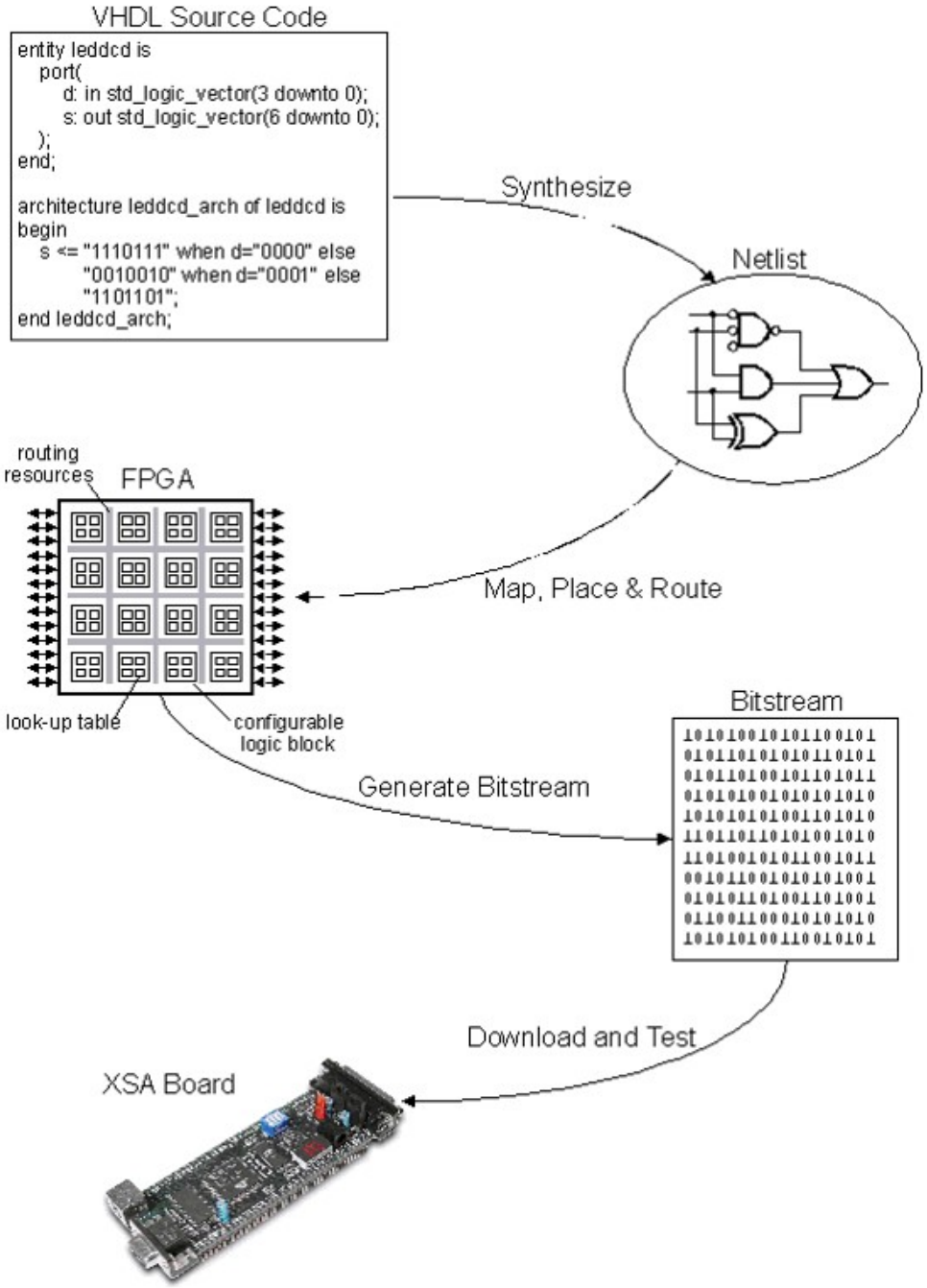
What are FGPAs

Each logic block has (at least):

- a 4- or 6-input LUT
- 1 flip-flop
- a mux for selecting combinatorial or registered output



Logic blocks can be interconnected in different ways


VHDL Source Code
entity leddcd is
port(
d: in std_logic_vector(3 downto 0);
s: out std_logic_vector(6 downto 0);
);
end;
architecture leddcd_arch of leddcd is
begin
s <= "1110111" when d="0000" else
"0010010" when d="0001" else
"1101101";
end leddcd_arch;
Synthesize
Netlist
routing resources
FPGA
look-up table
configurable logic block
Map, Place & Route
Bitstream
Generate Bitstream
Download and Test
XSA Board

FPGAs PROs and CONs

Main pros

- Lower design costs and lower design risks than with ASICs/ASSPs.
- Flexibility.
- Highly parallel processing (hundreds or thousands of “operations” executed simultaneously).
- High level of SoC (system-on-chip) integration (built-in features can reduce the number of components on the board).
- High density (top FPGAs have 200000 six-input logic functions in one chip).
- High performance (modern FPGAs operate on frequencies of as high as 550 MHz i.e. only few times less than typical custom ASIC top frequency).

FPGAs PROs and CONs

Main cons

- Relatively power-hungry devices
- Higher costs (from around one hundred to thousands dollars) than ASICs/ASSPs for high-volume production
- For “simple” applications lower-cost devices do exist
- For control-oriented applications MCUs and DSPs may be better solutions

Conclusion on CPLDs and FPGAs

FPGA contains up to 100,000 of tiny logic blocks while CPLD contains only a few blocks of logic that reaches up to a few thousands.

In terms of architecture, FPGAs are considered as ‘fine-grain’ devices while CPLDs are ‘coarse-grain’.

FPGAs are great for more complex applications while CPLDs are better for simpler ones.

FPGA is a RAM-based digital logic chip while CPLD is EEPROM-based.

Normally, FPGAs are more expensive while CPLDs are much cheaper.

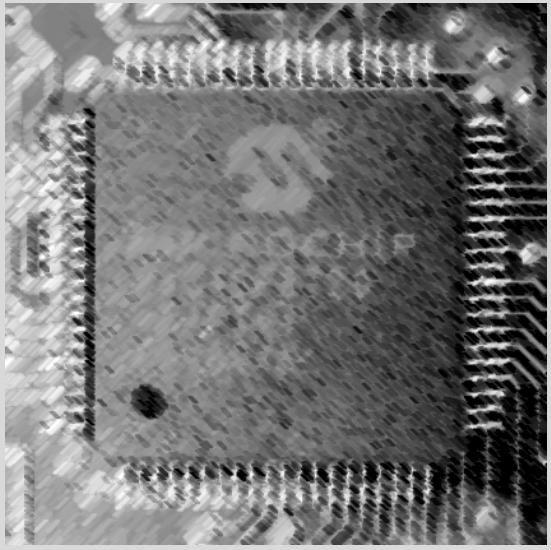
Delays are much more predictable in CPLDs than in FPGAs.

Use CPLDs, for small designs, where "instant-on", fast and wide decoding, ultra-low idle power consumption, and design security are important (e.g., in battery-operated equipment).

Use FPGAs for larger and more complex designs.

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MCUs and DSPs



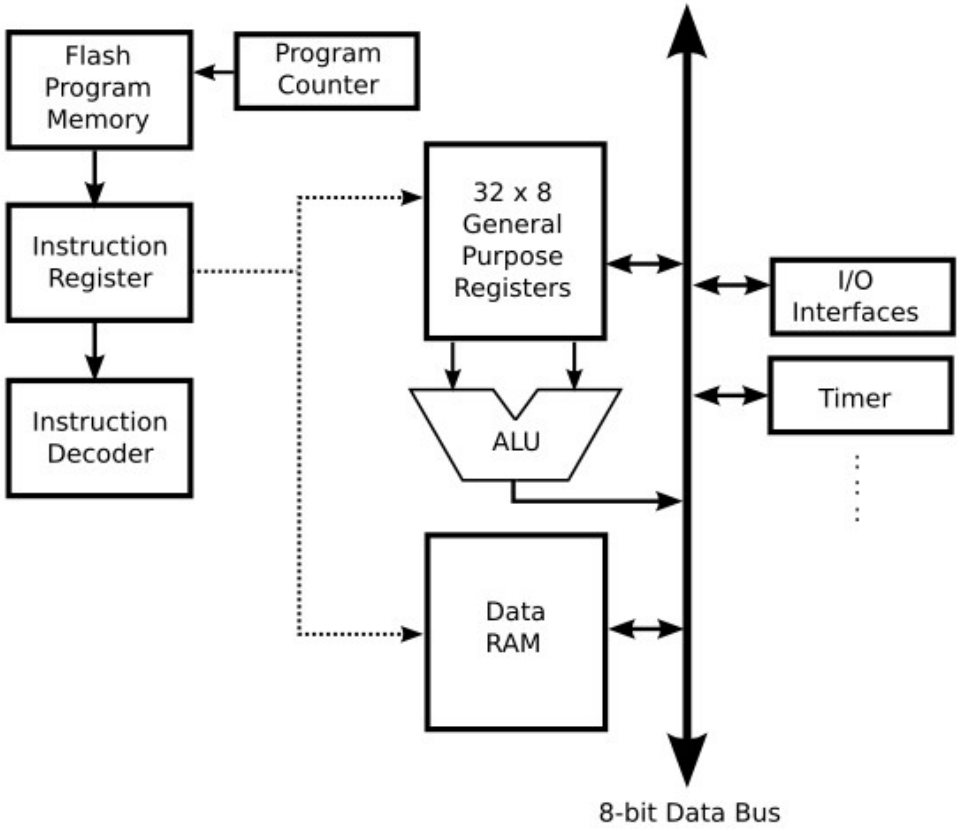
What are MCUs

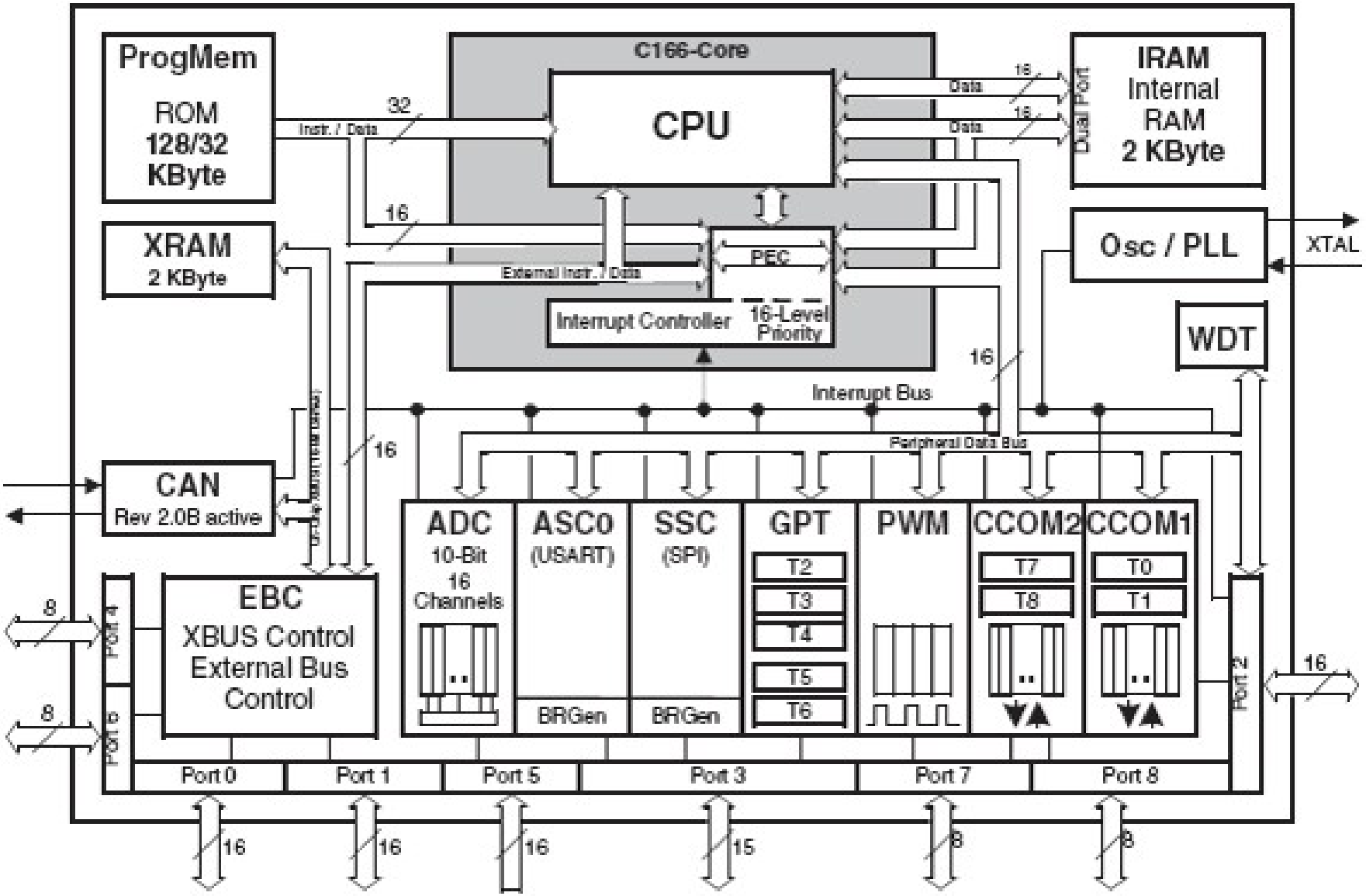
MCU (MicroController Unit): a single-chip microcomputer; i.e. a small computer on a single IC consisting of a relatively simple CPU combined with internal memory blocks and I/O control units.

Main Features:

- Unlike PLDs, MCUs have fixed wiring (i.e. the connections between its constituent gates are fixed)
- MCUs are sequentially programmable devices (i.e. they are re-programmed by changing a sequence of fixed operations, instructions, stored in memory)
- In contrast to general-purpose microprocessors, MCUs are simpler, lower-cost, lower-size and lower power consuming devices.
- Microcontrollers are primarily used in control-oriented applications (often interrupt-driven), for sensing and controlling external events

8-bit MCU


Flash Program Memory
Program Counter
Instruction Register
Instruction Decoder
32 x 8 General Purpose Registers
ALU
Data RAM
I/O Interfaces
Timer
8-bit Data Bus


ProgMem
ROM
128/32
KByte
32
Instr. / Data
C166-Core
CPU
16
Data
16
Data
Dual Port
IRAM
Internal
RAM
2 KByte
16
XRAM
2 KByte
PEC
External Instr. / Data
Interrupt Controller
16-Level
Priority
Osc / PLL
XTAL
WDT
16
Interrupt Bus
Peripheral Data Bus
16
CAN
Rev 2.0B active
ADC
10-Bit
16
Channels
ASC0
(USART)
SSC
(SPI)
GPT
T2
T3
T4
T5
T6
PWM
CCOM2
T7
T8
CCOM1
T0
T1
BRGen
BRGen
8
Port 4
8
Port 6
EBC
XBUS Control
External Bus
Control
Port 2
16
Port 0
Port 1
Port 5
Port 3
Port 7
Port 8
16
16
16
15
8
8

Typical MCUs Resources

CPU core (ranging from small and simple 4-bit processors to complex 32- or 64-bit processors)

CPU registers and SFRs (Special Function Registers) for controlling other peripherals

volatile memory (RAM) for data storage

non-volatile memory (ROM, EPROM, EEPROM or Flash memory) for program storage

programmable GPIO pins and special functions pins

serial I/O such as serial ports (UARTs)

other serial communications interfaces like I^2C, SPI and CAN

timers

PWM generators

watchdog

clock generator circuitry (driven by internal oscillators or external crystals)

ADC and DAC

in-circuit programming and debugging support

Programming MCUs

MCUs typically do not have a FPU or a math coprocessor, so both fixed point and floating-point arithmetic are performed by program code.

Peripherals programming skills are required (i.e. a deeper hardware knowledge than what needed for desktop applications)

Event-driven programming skills

MCUs were originally programmed only in assembly language.

Various high-level programming languages (like C) are now also in common use to target microcontrollers.

High-level compilers typically have some restrictions as well as enhancements to better support the unique characteristics of microcontrollers.

An MCU instruction set usually has many instructions intended for bit-wise operations. (For example, a general-purpose processor might require several low-level instructions to test a bit in a register and branch if the bit is set, where a MCU could have a single instruction to provide that commonly-required function.)

What are DSPs

DSP (Digital Signal Processor): a type of fast and powerful microprocessor whose architecture is optimized for performing intense “number crunching” applications

Main Features:

- Like MCUs, DSPs have fixed wiring (i.e. the connections between its constituent gates are fixed)
- DSPs are sequentially programmable devices (i.e. they are re-programmed by changing a sequence of fixed operations, instructions, stored in memory)
- In contrast to MCUs, DSPs have an architecture optimized for intensive number crunching operations.
- DSPs have typically fewer on-chip peripherals than MCUs.
- DSPs are primarily used in systems that require the precision processing of analog or digital signals.

DSPs Evolution

Original DSP ICs had a very simple structure with a powerful core and few additional on-chip peripherals



Modern DSP ICs had more and more peripherals and improved programmability

DSPs Peculiarities

fast multipliers (dedicated circuitry for MAC (multiply-accumulation) operations)

multiple execution units (such as ALUs working in parallel with MAC units)

special SIMD (single instruction, multiple data) operations

circular buffers

extended instruction sets

efficient memory access

- Harvard or modified-Harvard architecture
- more data-buses

dedicated AGU (Address Generation Units)

few-bit data format (16 to 32 bit architecture with accumulator registers)

efficient looping

streamlined I/O (low-latency interrupt, on-chip DMAs)

deep instruction pipeline

support for floating point operations (optionally)

multicore (optionally)

DSPs Applications

cell phones (according to Texas Instruments, DSPs are used as the engine in 70% of the world's digital cellular phones)

fax machines, high-resolution printers

home audio equipment, high fidelity sound

video game consoles

car (the anti-lock braking system)

computer disk drives

digital radios

digital cameras

audio and speech signal processing

sonar and radar signal processing

sensor array processing

spectral estimation

statistical signal processing

3D digital image processing

signal processing for communications

scientific simulations

biomedical signal processing

Conclusion on MCU vs DSP

MCU (Microcontroller)

- General-purpose embedded controller
- Rich peripherals (ADC, UART, SPI, CAN, timers)
- Good for control logic + system management
- Lower cost and power
- Suitable for kHz-range control loops

DSP (Digital Signal Processor)

- Optimized for numerical computation
- Fast MAC units, SIMD, Harvard architecture
- Designed for filtering, FFT, motor control, power electronics
- Deterministic high-frequency loops
- Higher computational throughput

Conclusion on MCU vs DSP

Design Rule of Thumb

If the bottleneck is **system integration → MCU**

If the bottleneck is **math throughput → DSP**



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